



Semiconductor Industry Development in State of Chihuahua

A proposal from



Presented to



December 2025

ASU-Chihuahua Semiconductor Faculty Development Institute

Introduction

The Ira A Fulton Schools of Engineering is pleased to present this workforce development proposal to the Mexican State of Chihuahua, Mexico, resulting from discussions between Arizona State University (ASU) and the Secretaría de Innovación y Desarrollo Económico (SIDE) in August 2025.

Chihuahua

Strategic Assets for Semiconductor Investment

With over 18,286. engineers and a strong manufacturing base, Chihuahua leads in aerospace, automotive, and electronics. Its export-driven economy and the new Taiwanese Industrial Park reinforce its role as a strategic link to U.S. and global markets.

Strategic Advantages

- Strong U.S. integration with logistics corridors across Texas for chip exports/imports.
- Active workforce programs (e.g. Taiwan training initiative) focus on semis, EV, automation

Competitiveness

Chihuahua ranks **8th out of 32 states** in the 2025 State Competitiveness Index.

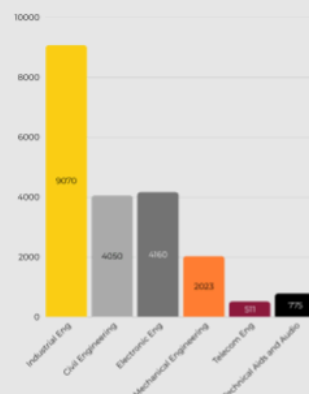
It performs best in the following areas:

- **1st** in exports performance
- **4th** in informality rate
- **4th** in economic complexity

Source: IMCO 2025

Engineering Workforce (Q3 2024)

Total: 18,286



Source: ENOE / Data México

FDI (199-2024)

- **United States:** US \$ 21.1 B
- **Canada:** US \$ 5.26 B
- **Ireland:** US \$ 3.08 B

Source: ENOE / Data México

Workforce Distribution by Sector



Source: ENOE / Data México

Strategic Companies



With 18,000+ engineers and a strategic U.S. border location, Chihuahua is a key industrial hub in advanced manufacturing, electronics, aerospace, and medical devices—powered by strong logistics and cross-border trade. As the home to strong industry partners including Foxconn, Jabil, Siemens and a specialized Taiwanese Industrial park, the state is in a prime position to grow its role in the semiconductor ecosystem.

The aim of this proposal is to build upon the knowledge acquired in prior development programs and deepen the expertise of program participants across Chihuahua, Mexico, within specific semiconductor-related priorities. ASU is dedicated to collaborating to align training programs with the specific technical development needs to support workforce development across the state of Chihuahua.

The program below provides a high-level estimate that will be built upon and customized as we align intended learning outcomes. To maximize high-impact program delivery, participants will be grouped into cohorts related to semiconductor design, advanced manufacturing, quality, and software development/automation as they relate to the semiconductor ecosystem.

1) ASU Semiconductor Strategy Co-Design and Immersion ATP Program

Ensuring alignment of learning outcomes alongside industry partners is paramount for this program. To achieve this, ASU will implement a structured four-week co-design process in coordination with the Secretaría de Innovación y Desarrollo Económico of Chihuahua.

Over the course of **four weekly sessions**, ASU faculty experts will meet with a designated **Co-Design Group** to define workforce development needs, map industry priorities, and match them with ASU's academic portfolio. The goal of this process is to collaboratively design a program that directly addresses the talent requirements of Chihuahua's industry sectors.

The **Co-Design Group** will be convened by SE and will include:

- **Industry specialists** with a direct understanding of current and emerging workforce needs.
- **Local educational ecosystem representatives** to ensure academic alignment.
- **Government officials** to validate policy relevance and facilitate implementation.

This participatory approach will ensure that the program reflects local realities, integrates the perspectives of multiple stakeholders, and provides a tailored solution to support Chihuahua's economic development goals.

Additionally, ASU may engage prospective participants who have completed prior training programs to assess previously covered content, ensuring a coherent sequence of learning and avoiding overlap.

Program Options

This program can include four flexible programming options, organized and led by ASU to align with the needs of faculty participants from across Chihuahua. Each option varies in duration and depth of engagement, allowing partners to select the format that best supports their institutional goals. The four options include:

- **5-Month Online Program:** Two live online sessions per month with ASU faculty and industry experts focused on the priority topics, with learning outcomes decided upon via the co-design process prior to launch.

Sample Program - Semiconductor Assembly, Test, and Packaging Track

This track supports faculty in disciplines who are ready to advance beyond foundational semiconductor concepts and deepen their understanding of backend manufacturing. The focus is on packaging flows, assembly methods, semiconductor test, and reliability practices that align with the types of operations present or emerging in Chihuahua's industrial ecosystem.

Participants will explore core ATP processes, understand how backend steps connect to system-level manufacturing, and gain clarity on which competencies are most relevant for preparing students for roles in local industry. Weekly sessions include two hours of technical instruction and two hours of structured industry insights drawn from real ATP environments, manufacturing partners in the region, and global best practices.

The goal is to strengthen faculty ability to integrate ATP-relevant concepts into coursework without requiring specialized lab facilities or equipment.

By the end of this program, faculty participants will be able to:

1. Explain how assembly, test, and packaging fit into the semiconductor value chain and how these processes connect to manufacturing activities common in Chihuahua.
- 2.

Module	Topics	Week 1	Week 2	Week 3	Week 4	Week 5
Month 1 Overview of Backend Manufacturing and Packaging Flows	• How assembly, test, and packaging fit into the semiconductor value chain • Common packaging workflows used in cost-sensitive, high-volume manufacturing • Processes commonly seen in Chihuahua's electronics and EMS sectors • Key materials and equipment used in backend operations • Practical ways to introduce packaging concepts in classroom settings	2hr		2hr		
		Asynchronous Course and Resources				

Month 2 Assembly Fundamentals and Interconnect Techniques	• Die attach basics and factors that influence reliability • Wire bonding and its continued relevance in regional industries • When and why manufacturers transition to flip-chip or advanced interconnects • Substrates, leadframes, and SMT operations that bridge chips to system-level products • Common defects introduced during assembly and how to teach process control concepts	2hr		2hr		
Asynchronous Course and Resources						
Month 3 Introduction to Semiconductor Test and Automated Test Methods	• Where and why semiconductor test occurs • Final test workflows that align with electronics manufacturing capabilities in Chihuahua • Basic automated test concepts and accessible tester architectures • Functional testing approaches relevant for PCB-level and system-level products • Using test data to reinforce classroom lessons on yield, variation, and reliability	2hr		2hr		
Asynchronous Course and Resources						
Month 4 Reliability, Quality Systems, and Failure Analysis Essentials	• Reliability drivers for package and assembly integrity • High-level failure analysis methods that can be taught without specialized tools • Quality practices used by automotive, aerospace, and medical device suppliers in Chihuahua • SPC, FMEA, and other frameworks for teaching quality and continuous improvement • How to weave reliability concepts into labs, projects, and capstones	2hr		2hr		
Asynchronous Course and Resources						
Month 5 Emerging Packaging Trends and Curriculum Application	• High-level overview of advanced packaging directions such as SiP and heterogeneous integration • Realistic workforce opportunities for Chihuahua based on current and near-term capabilities • How automation and data analytics are affecting ATP roles	2hr		2hr		

	• Strategies for engaging local industry to validate content and create applied student experiences	Asynchronous Course and Resources
Plus the Semiconductor Packaging Immersion In-person Tempe, AZ		

- Option 1) 3-day Semiconductor Packaging Immersion:** A focused three-day in-person program at ASU's Tempe campus designed to provide targeted engagement, applied learning experiences, and foundational academic planning support. Visits include:
 - Three full days of structured sessions combining interactive workshops, guided discussions, and collaborative activities.
 - Roundtable engagements with ASU faculty and staff to support curriculum development and instructional planning.
 - Guided tours of ASU labs, teaching facilities, and research centers aligned with co-designed program themes.
 - Exposure to educational best practices, including strategies for workforce-aligned instruction and academic program design.
 - Small-group format (up to 25 participants per cohort) to maximize hands-on learning and peer exchange.
 - Flexibility to tailor content and activities based on priorities identified during the co-design process.
- Option 2) 5-day Semiconductor Packaging Immersion:** A week-long in-person experience hosted at ASU's Tempe campus designed to provide extended engagement, cross-disciplinary collaboration, and practical application aligned with program goals. Visits include:
 - Five full days of in-person sessions combining technical exploration, teaching strategies, and academic program development.
 - Interactive workshops with ASU faculty focused on practical application, lab exposure, and education design best practices.
 - Extended time for peer collaboration and guided curriculum planning to support implementation in home institutions.
 - Site visits to ASU labs, innovation centers, and partner facilities to contextualize learning and connect with industry environments.
 - Faculty roundtables and strategy sessions focused on educational alignment and workforce readiness.
 - Flexibility to tailor sessions based on outcomes from the co-design process

- **Option 3) 2-Week In-Person Immersion:** A comprehensive two-week in-person program at ASU that allows for deeper engagement, applied project work, and cross-cultural academic exchange. Visits include:
 - Ten full days of immersive learning, with dedicated time for structured workshops, hands-on activities, and collaborative curriculum development.
 - Opportunities for applied project work or prototype development aligned with institutional goals.
 - Engagement with ASU instructional design teams for in-depth support on course or program creation.
 - Cultural enrichment experiences and networking activities to foster long-term academic partnerships.
 - On-site visits to research facilities, teaching labs, and relevant innovation hubs.
 - Built-in flexibility to align the program flow with themes and outcomes defined during the co-design phase.

Asynchronous Courses and Resources

- **Arizona Semiconductor Story:** Participants will be provided with access to ASU's exclusive Arizona Semiconductor Story 6-module course, providing insight into how Arizona became known as the "Silicon Desert," building a thriving semiconductor ecosystem across the full value chain from fabrication to packaging to innovation.



The Arizona Story
A Case Study in Ecosystem Building

What It Is:
A six-part case study embedded in the Master Class, showcasing how Arizona built a thriving semiconductor ecosystem across the full value chain from fabrication to packaging to innovation.

Why It Matters:

- Grounds technical content in a real-world success story
- Highlights how policy, investment, and partnerships drive impact
- Offers a replicable model for regional ecosystem growth

Faculty Role:

- Record short video clips or recommend SMEs to share story-driven insights
- Help connect weekly themes to learners' local or institutional context through weekly async assignments
- Suggest media or guest voices that enrich the case study discussion

Arizona Semiconductor Story:
A case Study in Ecosystem Building
[Start Course](#)

- **Best Practices for Teaching and Learning** Participants will have access to resources developed by expert learning experience designers to support innovative models and techniques, including flipped classroom, engagement techniques, inclusive and supporting learning environments, and more.

ASU Core Facilities

Throughout the immersion programs, participants will have opportunities to visit some of ASU's state-of-the-art research facilities, which support advanced semiconductor research,

prototyping, and workforce training. These facilities are instrumental in driving innovation and industry collaboration.

ASU MacroTechnology Works

ASU's MacroTechnology Works offers industry-scale capabilities for advanced electronics, photonics, and solar fabrication. These facilities support secure prototyping, testing, and production of new technologies and play a central role in ASU's Advanced Materials, Processes and Energy Devices (AMPED) Science and Technology Center. They also drive ASU's semiconductor research and workforce development efforts.

ASU NanoFab

ASU NanoFab is a state-of-the-art nanofabrication and characterization facility supporting university research and industry prototype development. Expertise includes nanofabrication, silicon processing, bioelectronics, MEMS, nano-fluidics, and optoelectronics. Experienced staff provide technical support.

Eyring Materials Center

The Eyring Materials Center offers open-access facilities for materials characterization, surface analysis, and high-resolution electron microscopy across scientific disciplines. It includes:

- **John M. Cowley Center** – A leader in high-resolution electron microscopy, housing a dozen microscopes for atomic-scale material analysis.
- **Life Science Electron Microscopy** – A specialized lab for biological EM analysis, sample processing, and high-pressure freezing.
- **Goldwater Materials Science Facility** – Equipped for optical, structural, and surface analysis, including X-ray diffraction, atomic force microscopy, and spectroscopy techniques.

ASU Certificate of Completion



Upon program completion, participants will receive a Certificate of Completion from ASU's Ira A. Fulton Schools of Engineering. As one of the world's top-ranked engineering schools, an ASU certification is globally recognized.

Program Investment

The table below outlines the four available program format options, each with corresponding levels of engagement and pricing. Program fees cover instruction by ASU faculty and experts, curated learning experiences, reference materials, and program coordination. Participants who complete any of the immersion formats will receive a Certificate of Completion from the Ira A. Fulton Schools of Engineering at Arizona State University.

Option	Depth of Engagement	Key Value Drivers	Program Fee and Optional Participant Travel	Per Cohort (25 members)
5-Month Online Program	Foundational Overview + Topic Exposure	Live online sessions with ASU faculty and experts; flexible learning; baseline knowledge across key themes co-designed with stakeholders	Program Fee: \$2,350	Program Fee: \$58,750 Total Cost: \$58,750
Plus one of the next 3 alternatives in site immersive models				
Option 1) 3-Day In-Person Immersion	Strategic Focus + Best Practices	Targeted sessions with experts; roundtables and lab tours; exposure to ASU models for workforce-aligned education; rapid engagement for experienced participants	Program Fee: \$1950 + Participant Travel: \$2,050	Program Fee: \$48,750 + Participant Travel: \$51,250 Total Cost: \$158,700
Option 2) 5-Day In-Person Immersion	Structured Exploration + Collaboration	Extended hands-on sessions; greater interaction with ASU faculty; time for peer exchange and planning; foundational academic development	Program Fee: \$3,500 + Participant Travel: \$2,600	Program Fee: \$87,500 + Participant Travel: \$65,000 Total Cost: \$211,250
Option 3) 2-Week In-Person Immersion	Project-Based Development + Co-Creation	Team-based curriculum or strategy development; instructional design support; applied activities; cultural and institutional immersion	Program Fee: \$5,000 + Participant Travel: \$4,200	Program Fee: \$137,500 + Participant Travel: \$105,000 Total Cost: \$301,250

2) Participation in the ASU Semiconductor Materials Immersion Program

ASU will offer 30 participation slots for Mexican students in the **Semiconductor Materials Immersion Program**. This program is a four-week blended experience that combines two weeks of online learning with two weeks of in-person immersion at ASU in Tempe, Arizona. It is designed to provide participants with both theoretical knowledge and practical skills in semiconductor materials science, preparing them for careers in microelectronics and advanced manufacturing.

The program begins with self-paced and live online sessions covering English for Semiconductors, industry foundations, safety protocols, and introductory materials science. It then transitions to a two-week in-person immersion at ASU, where students engage in advanced coursework on alloys, diffusion, polymers, ceramics, and semiconductor devices, complemented by lab tours, hands-on experiences in state-of-the-art facilities, and panels with industry experts.

Participants will also complete a capstone project in which they research an advanced semiconductor application and present their findings in both written and oral formats. Upon completion, students receive an official certificate from ASU's Ira A. Fulton Schools of Engineering as well as a digital microcredential in Semiconductor Materials Science.

Beyond the classroom, the program includes cultural enrichment activities such as a welcome reception, networking with ASU students and faculty, a guided tour of the Grand Canyon, and access to ASU's fitness and wellness facilities. Housing, meals, transportation, and health insurance are provided as part of the package, ensuring a fully supported and immersive experience.

Day	Topics
Semiconductor Introduction (Online, Self Paced)	
Self Paced: English for the Semiconductor Industry	• Introduction to Semiconductors • Semiconductor Design • Semiconductor Manufacturing • Equipment and Safety Protocols • Semiconductor Supply Chain • Industry relevant Human Resources and Finance
Semiconductor Materials I (Synchronous through Zoom)	
Monday - Introduction to Materials	• Applications & relevance of materials science in semiconductor industry • Materials science

	• Semiconductors • Metals • Polymers • Ceramics
Tuesday - Bonding & Periodic Table	• Applications & relevance of bonding & periodic table in semiconductor industry • Atomic bonding in semiconductors ionic bonding • Covalent bonding, metallic bonding • Van Der Waals Bonding, hydrogen bonding • Periodic Table
Wednesday - Crystal Structure, unit cells, index directions	• Applications & relevance of crystal structure in semiconductor industry • Crystal structure in semiconductors • Unit cells • Index directions • Amorphous materials • Crystalline materials
Thursday - Crystal Structure, index planes, defects	• Applications & relevance of crystal structure & defects in semiconductor industry • Crystal index planes • Slip systems in semiconductors • Defects in semiconductors • Point defects in semiconductors • Line defects in semiconductors
Friday - Mechanical Properties, Tensile test, environment, failure	• Applications & relevance of mechanical properties in semiconductor industry • Mechanical properties of semiconductor materials • Mechanical testing of materials • Elastic deformation of materials • Plastic deformation of materials • Environmental effects on materials • Fracture & failure of materials
Semiconductor Materials II (In-Person at ASU)	
Monday - Alloys, Diffusion, Strength	• Applications & relevance of alloys & diffusion in semiconductor industry • Diffusion in semiconductor materials • Strength of materials • Plastic deformation • Annealing • Recrystallization
Tuesday - Phase	• Applications & relevance of phase diagrams and solubility in

diagrams, solubility, eutectic behavior	semiconductor industry • Phase diagrams • Solid solubility • Microstructure • Eutectic reactions • Eutectic structures
Wednesday - Phase diagram, model system, heat treatment	• Applications & relevance of heat treatment in semiconductor industry • Understanding a model phase diagram • Heat treatment • Phase transformation • Quenching • Isothermal Transformation diagram
Thursday - Age hardening, polymers, concepts, synthesis	• Applications & relevance of age-hardening, and polymer synthesis in semiconductor industry • Age-hardening • Nucleation & crystal-growth • Precipitation • Polymer synthesis • Physical characterization of polymers
Friday - Project Kickoff	• Start project (Research an advanced semiconductor application and write a summary paper and presentation).
Monday - Polymers, processing, thermo-mechanical properties	• Applications & Relevance of polymer processing in semiconductor industry • Polymer processing • Polymer molding & extrusion • Glass-transition temperature • Log-modulus vs temperature curve • Thermal properties • Mechanical Properties
Tuesday - Conductivity, electrical behavior, semiconductor devices	• Applications & Relevance of electrical and semiconductor devices in semiconductor industry • Semiconductor devices • Band-Structure of Semiconductors • Elemental & Compound Semiconductors • Intrinsic & Extrinsic Semiconductors • Lab Tour of Eyring Materials Center, and NanoFab
Wednesday - Ceramics, structure, properties, processing, applications, composites	• Applications & Relevance of ceramics and composites in semiconductor industry. • Overview of Ceramics & Composites • Structure of Ceramics & Composites • Properties of Ceramics & Composites

	• Network Modifiers • Strengthening of Ceramics & Composites
Thursday - Industry Day and Project Work	• Lab Visit to MacroTechnology Works, Advanced Electronics and Photonics Lab, and Solar Power Fab. • Work on Project (Research an advanced semiconductor application and write a summary paper and presentation).
Friday - Project Showcase and Closing Ceremonies	• Completion of project (Research an advanced semiconductor application and write a summary paper and presentation).

**Note days may change for site and lab visits pending availability at the time of the program.*

Through this program, participants will gain:

- A strong foundation in semiconductor materials and their device applications.
- Hands-on exposure to advanced labs and tools, linking theory with practice.
- Improved technical English for the semiconductor industry.
- Insights from U.S. industry leaders and academic experts.
- International certification that strengthens employability in the semiconductor sector.

Program Investment

Program fees cover instruction by ASU faculty and experts, curated learning experiences, reference materials, and program coordination. In addition to the program fee, optional participant travel costs are listed for the in-person sessions in ASU.

Participants who complete any of the immersion formats will receive a Certificate of Completion from the Ira A. Fulton Schools of Engineering at Arizona State University.

Item	Cost Per Person	For 30 Students
Migratory Document Fees	\$400	\$ 12,000
Course fees	\$588	\$ 17,640
Books and materials	\$80	\$ 2,400
On-campus or nearby housing	\$954	\$ 28,620
Three daily meals on-campus or nearby	\$480	\$ 14,400

Transportation within campus and to and from the nearest airport	\$80	\$ 2,400
Medical insurance for the course of the in-person program	\$96	\$ 2,880
Cultural activities and trips	\$305	\$ 9,150
Round-trip airfare from Mexico City or Chihuahua (CUU) to the host city (Optional)	\$1,033	\$ 30,990
Administrative cost of the issuing of a diploma or certificate of completion at the end of the program	\$20	\$ 600
Administrative costs	\$964	\$28,920
		\$ 150,000

Total Cost per student: USD \$5,000

USD \$5,000 per participant for a cohort of 30 students, inclusive of online and in-person instruction, housing, meals, airfare, cultural activities, transportation, insurance, and certification.

3) Semiconductor Ecosystem Master Class

ASU will offer 20 participation slots for the Chihuahua's selected faculty, government officials, and professionals in the Semiconductor Ecosystem Master Class. This Master Class is a six-week, live and synchronous program led by ASU faculty and industry leaders. It follows a structured progression that builds from fundamentals to application, with two 90-minute sessions per week, and covers the global semiconductor value chain, including industry foundations, supply-chain dynamics, chip design and fabrication, advanced packaging, assembly and test, and emerging trends. The program culminates in a capstone assignment called the Semiconductor Ecosystem Action Plan, and participants receive a certificate of completion from the Ira A. Fulton Schools of Engineering.

For the State of Mexico's engagement, the six designated decision makers will participate as a dedicated cohort. Over the six weeks, they will complete short, structured weekly deliverables designed to act as catalysts for the state strategy, such

as opportunity scans tied to the pre-feasibility findings, stakeholder and asset maps, initial curriculum alignment notes, and policy or workforce concept briefs. These assignments will receive personalized feedback from ASU experts and targeted coaching to ensure that outputs are practical, decision-ready, and directly feed the broader program elements, including the pre-feasibility assessment, the Arizona study tour, and the MoU.

Participants will also study the Arizona Semiconductor Story as a documentary-style case, using weekly themes to distill best practices in policy, workforce development, investment attraction, and ecosystem governance that can inform the State of Mexico's strategy. In this tailored format, the Master Class remains scalable, structured, and cost-effective while blending expert-led instruction, industry engagement, and hands-on work products that translate learning into an actionable state strategy

Through the Master Class, participants will gain foundational knowledge across critical areas such as supply chain dynamics, chip design, manufacturing processes, and workforce development. The main goals of this activity are to:

- Build a shared baseline of semiconductor fundamentals for the Chihuahua's core cohort of decision makers, faculty, and officials to inform the state-level strategy.
- Translate learning into action through weekly assignments that accumulate into a draft strategy for the Chihuahua, including opportunity mapping, curriculum and workforce alignment, and policy options, with personalized feedback from ASU experts.
- Expand strategic networks by connecting participants with industry leaders and academic experts during live sessions and Q&A, creating pathways for future collaborations and talent initiatives.
- Align and de-risk next steps by using cohort outputs to shape the Arizona study tour agenda and the scope of the MoU, establishing clear Year-1 priorities, governance, and implementation milestones.

Semiconductor Ecosystem Master Class

Executive 6-week online synchronous program

Weekly Themes

-  Introduction to Semiconductors: History, Basics, and Applications
-  Global Semiconductor Supply Chain
-  Semiconductor Design and Fabrication
-  Semiconductor Packaging
-  Semiconductor Assembly and Test
-  Emerging Trends and Opportunities

+2 weeks of Curriculum Development

During this stage, faculty will draft their Curriculum Action Plan to guide implementation at their institution.

Weekly Schedule

Sample structure of a week during the program

	M	T	W	TH	F
Synchronous Sessions	Technical Session with ASU Expert 90 min.		Industry Expert Session 90 min.		
Asynchronous Work	The Arizona Story: A Case Study in Ecosystem Building Explore how Arizona became one of the United States' most prominent semiconductor hubs. Through six thematic segments, the case study examines key decisions, investments, and partnerships that transformed the state into a model for ecosystem development				
Weekly Program Deliverables	Ecosystem Strategy (For all Participants) Create a strategic document for actionable opportunities in growth, innovation, and investment in the semiconductor sector.		Curriculum Action Plan (For faculty participants only) Develop a curriculum action plan outlining new or enhanced courses aligned with the semiconductor industry's needs and best practices.		

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Program Investment

Program fees cover instruction by ASU faculty and experts, curated learning experiences, reference materials, and program coordination.

Participants who complete the program will receive a Certificate of Completion from the Ira A. Fulton Schools of Engineering at Arizona State University.

Total Cost: \$35,200

(USD 1,760 per participant)

For a group of up to 20 participants enrolled in the Semiconductor Ecosystem Master Class. (\$1,760* per person, next cohort to be announced). This pricing reflects a 20% discount applicable to groups of 20 or more.

4. Budget

Itemized Budget

Program Name	Duration	Format	Size # Participants	Per Participant Cost	Total Cost Travel included	Key Features
ASU Semiconductor Strategy Co-Design and Immersion Program	5 months (online) + Optional in-person immersion	Online + Optional In-Person	25	\$2,350	\$58,750 (online) + 3 day option with travel \$100,000 (3-day) Total \$158,700 or 5-day option with travel \$152,500 Total \$ 211,250 or 2 week option with travel \$242,500 Total: \$301,250	Includes co-design phase with stakeholders; two live sessions per month with ASU faculty and experts; immersion options in Tempe focused on strategic alignment, collaboration, or co-creation.
	4 weeks (2 online + 2 in-person)	Blended (Online + In-Person)	30	\$5,000	\$150,000 Or \$120,000 (this pricing does not include travels fees*)	Comprehensive program combining theory and practice in semiconductor materials science; includes housing, meals, airfare, cultural activities, local transport, insurance, and certification.
	Semiconductor Ecosystem Master Class	Online	20	\$1,760	\$35,200	Live sessions with ASU faculty and industry mentors; includes Arizona case study and strategic deliverables; 20% group discount applied.

Budget Narrative

- **ASU Semiconductor Strategy Co-Design and Immersion Program**

USD \$52,500 for a five-month online program for up to 25 participants (USD \$2,100 per person) + Immersive experience, featuring two live sessions per month with ASU faculty and industry experts. The program begins with a structured co-design phase alongside stakeholders to define local priorities and learning outcomes, ensuring that each module aligns with industry needs and regional strategies. Participants may also extend their engagement through an in-person immersion at ASU in Tempe, Arizona, with three available options: a **3-day** program focused on strategic alignment and best practices (Total Cost: USD \$98,750), a **5-day immersion** emphasizing collaboration and applied learning (Total Cost: USD \$150,000), or a **2-week co-creation residency** for project-based curriculum or strategy development (Total Cost: USD \$230,000).

- **ASU Semiconductor Materials Immersion Program**

USD \$150,000 for a cohort of up to 30 students (USD \$5,000 per participant). This Program is a four-week blended experience combining two weeks of online learning with two weeks of in-person immersion at ASU in Tempe, Arizona. The total cost includes instruction, housing, meals, airfare (optional), cultural activities, local transportation, insurance, and certification. This program equips participants with theoretical and practical foundations in semiconductor materials science through coursework, lab experiences, and engagement with industry experts.

- **Semiconductor Ecosystem Master Class**

USD \$35,200 for a group of up to 20 participants (USD \$1,760 per person). Includes enrollment in the Semiconductor Ecosystem Master Class, 2025, a six-week live and synchronous program featuring ASU faculty, the Arizona case study, and strategic mentorship for semiconductor ecosystem development. A 20% group discount has been applied, and additional participants may join at the same discounted per-person rate.

** Travel expenses include round trip airfare, lodging, meals and incidentals, and ground transportation. Rates used to calculate these expenses are in line with foreign per diem rates published by the U.S. Department of State. https://aoprals.state.gov/web920/per_diem.asp*

Contact Us

Our team at ASU Fulton Schools of Engineering Global Outreach and Extended Education office will support you every step of the way, from initial proposals and agreements to logistics,

faculty communications, billing, and additional support to ensure a smooth and effective program delivery.

To get started, ask questions, or make revisions to the proposed program, please contact Brittany Holmes, Associate Director for Professional Programs and Workforce Development at brittany.a.holmes@asu.edu.